Notes lab 3

(might contain typos)

October 25, 2024

Hi, person; during lab 3, we spent most of the time discussing a formula on the whiteboard, and even though I recorded the lab, I did not record the whiteboard, so now I'll briefly describe what we discussed.
It all started with Policy Iteration :

Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization
   $V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$

2. Policy Evaluation
   Loop:
   $\quad\Delta \leftarrow 0$
   $\quad$Loop for each $s \in \mathcal{S}$:
   $\quad\quad v \leftarrow V(s)$
   $\quad\quad V(s) \leftarrow \sum_{s',r} p(s', r \,|\, s, \pi(s))\big[r + \gamma V(s')\big]$
   $\quad\quad \Delta \leftarrow \max(\Delta, |v - V(s)|)$
   until $\Delta < \theta$ (a small positive number determining the accuracy of estimation)

3. Policy Improvement
   policy-stable $\leftarrow$ *true*
   For each $s \in \mathcal{S}$:
   $\quad$*old-action* $\leftarrow \pi(s)$
   $\quad\pi(s) \leftarrow \operatorname{argmax}_a \sum_{s',r} p(s', r \,|\, s, a)\big[r + \gamma V(s')\big]$
   $\quad$If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*
   If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

And specifically, the equation in it, and ideally, we wanted to get a grasp of the meaning or possibly a different (more "intuitive") explanation:

$$V(s) := \sum_{s',r} p(s', r|s, \pi(s)) \left[r + \gamma V(s')\right]$$

To begin with, we changed the notation for that policy expectation

$$\begin{aligned} V(s) :&= \sum_{s',r} p(s', r|s, \pi(s)) \left[r + \gamma V(s')\right] \\ &= \underbrace{\sum_{a} \pi(a|s)}_{\text{new stuff}} \sum_{s',r} p(s', r|s, a) \left[r + \gamma V(s')\right] \end{aligned}$$

Then, we realized that the right-hand side of this equation is the return:

$$[r + \gamma V(s')] = r + \gamma \mathbb{E}\left[\sum_{i=0} \gamma^i r_i\right] = \mathbb{E}\left[\sum_{j=0} \gamma^j r_j\right] = G_t$$

However, this notation is not very clear, as it does not explicitly tell what that expectation is over (are we averaging with respect to what?).
For this reason, from now on we will use this notation:

$$\mathbb{E}_{x \sim X}[f(x)]$$

This clearly states that we are taking random samples x from X, and then calculating $f(x)$.

At that point, we tried to understand the left-hand side. In particular, it contains a joint conditional distribution, $p(s', r|s, a)$, which might be scary at first glance.
First, recall from probability theory (chain rule of probability) that:

$$P(A, B) = P(A)P(B|A) = P(B)P(A|B)$$

The same holds true for conditional distributions:

$$P(A, B|C) = P(A|C)P(B|A, C) = P(B|C)P(A|B, C)$$

Therefore, we can factorize in two ways:

$$p(s', r|s, a) = p(s'|s, a)p(r|s, a, s') = p(r|s, a)p(s'|r, s, a)$$

Even though they are equivalent, they have completely different meanings. In particular, we will stick with the first one for the following reason

$$p(s', r|s, a) = \underbrace{p(s'|s, a)}_{\mathcal{P}}\underbrace{p(r|s, a, s')}_{\mathcal{R}}$$

where $\mathcal{P}$ is the transition matrix/distribution of the MDP (given the current state and the action you want to perform, it defines the set of states you can reach and with which probability), and $\mathcal{R}$ is the reward function (given the state you were, the action you performed, and the new state you ended up, it defines the reward and with which probability).
Given this, we can go on with our factorization

$$\begin{aligned}
V(s) :&= \sum_{s',r} p(s', r|s, \pi(s)) \left[r + \gamma V(s')\right] \\
&= \sum_{a} \pi(a|s) \sum_{s',r} p(s', r|s, a) \left[r + \gamma V(s')\right] \\
&= \sum_{a} \pi(a|s) \sum_{s',r} \underbrace{p(s'|s, a)p(r|s, a, s')}_{\text{new factorization}} \left[r + \gamma V(s')\right] \\
&= \sum_{a} \pi(a|s) \underbrace{\sum_{s'} \sum_{r}}_{\text{just separating the two summations}} p(s'|s, a)p(r|s, a, s') \left[r + \gamma V(s')\right]
\end{aligned}$$

Now, we can observe how the term $p(s'|s, a)$ does not depend on r (the second summation), therefore it's just a constant, and since $\sum cf(x) = c \sum f(x)$, we can just take it out:

$$\begin{aligned}
V(s) :&= \sum_{a} \pi(a|s) \sum_{s'} \sum_{r} p(s'|s, a)p(r|s, a, s') \left[r + \gamma V(s')\right] \\
&= \sum_{a} \pi(a|s) \underbrace{\sum_{s'} p(s'|s, a) \sum_{r}}_{\text{brought the term outside of the summation}} p(r|s, a, s') \left[r + \gamma V(s')\right]
\end{aligned}$$

The last step is to realize that $\mathbb{E}[f(x)] = \sum p(x)f(x)$; therefore, anything that looks like a sum-probability-function can be seen as an expectation over something.
Thanks to this, we can realize that the first summation with $\pi(a|s)$ is just an expectation:

$$\begin{aligned}
V(s) &:= \sum_{a} \pi(a|s) \sum_{s'} p(s'|s, a) \sum_{r} p(r|s, a, s') \left[r + \gamma V(s')\right] \\
&= \underbrace{\mathbb{E}_{a \sim \pi(a|s)}}_{\text{now we consider it as an expectation}} \left[\sum_{s'} p(s'|s, a) \sum_{r} p(r|s, a, s') \left[r + \gamma V(s')\right]\right]
\end{aligned}$$

But the same holds true for the summation with $p(s'|s,a)$:

$$\begin{aligned} V(s) &:= \mathbb{E}_{a\sim\pi(a|s)}\left[\sum_{s'} p(s'|s,a)\sum_r p(r|s,a,s')\left[r+\gamma V(s')\right]\right] \\ &= \mathbb{E}_{a\sim\pi(a|s)}\left[\mathbb{E}_{s'\sim p(s'|s,a)}\left[\sum_r p(r|s,a,s')\left[r+\gamma V(s')\right]\right]\right] \end{aligned}$$

And again, for the term with $p(r|s,a,s')$:

$$\begin{aligned} V(s) &:= \mathbb{E}_{a\sim\pi(a|s)}\left[\mathbb{E}_{s'\sim p(s'|s,a)}\left[\sum_r p(r|s,a,s')\left[r+\gamma V(s')\right]\right]\right] \\ &= \mathbb{E}_{a\sim\pi(a|s)}\left[\mathbb{E}_{s'\sim p(s'|s,a)}\left[\mathbb{E}_{r\sim p(r|s,a,s')}\left[r+\gamma V(s')\right]\right]\right] \end{aligned}$$

Now, this (imho) is much more readable, for the following reason. Let's denote the various steps of this equation:

$$V(s) = \underbrace{\mathbb{E}_{a\sim\pi(a|s)}}_{(1)}\left[\underbrace{\mathbb{E}_{s'\sim p(s'|s,a)}}_{(2)}\left[\underbrace{\mathbb{E}_{r\sim p(r|s,a,s')}}_{(3)}\left[r+\gamma V(s')\right]\right]\right]$$

We can interpret this in a very simple way:

> We are given a state s, then to estimate its value, I take an action according to $\pi(a|s)$ (1). Now that we have a state and an action, the MDP "teleports us to a new state" according to $p(s'|s,a)$ (2). Given that I was in a state, I took an action (1), I was teleported on a new state (2), I can get a reward according to $p(r|s,a,s')$ (3). Once we have all of this, the average of this whole procedure gives us the returns.

Earlier, we said that $G_t = \mathbb{E}\left[\sum_{j=t}\gamma^j r_j\right]$ was not so clear because we did not know what was that expectation taken over. However, we know that $G_t = V(s_t)$. So, say we are in s_0:

$$\begin{aligned} G_0 &= V(s_0) \\ &= \mathbb{E}_{a\sim\pi(a|s_0)}\left[\mathbb{E}_{s_1\sim p(s_1|s_0,a)}\left[\mathbb{E}_{r\sim p(r|s_0,a,s_1)}\left[r+\gamma V(s_1)\right]\right]\right] \end{aligned}$$

But, since we can recursively "unpack" the definition of $V(s)$ that we defined (the one with the three expectations), and we get the full definition of G_0:

$$\begin{aligned} G_0 &= V(s_0) \\ &= \mathbb{E}_{a_0\sim\pi(a_0|s_0)}\left[\mathbb{E}_{s_1\sim p(s_1|s_0,a_0)}\left[\mathbb{E}_{r_1\sim p(r_1|s_0,a_0,s_1)}\left[r_1+\gamma V(s_1)\right]\right]\right] \\ &= \mathbb{E}_{a_0\sim\pi(a_0|s_0)}\left[\mathbb{E}_{s_1\sim p(s_1|s_0,a_0)}\left[\mathbb{E}_{r_1\sim p(r|s_0,a_0,s_1)}\left[r_1+\gamma\mathbb{E}_{a_1\sim\pi(a_1|s_1)}\left[\mathbb{E}_{s_2\sim p(s_2|s_1,a_1)}\left[\mathbb{E}_{r_2\sim p(r_2|s_1,a_1,s_2)}\left[r_2+\gamma V(s_2)\right]\right]\right]\right]\right]\right] \\ &= \mathbb{E}_{a_0\sim\pi(a_0|s_0)}\left[\mathbb{E}_{s_1\sim p(s_1|s_0,a_0)}\left[\mathbb{E}_{r_1\sim p(r_1|s_0,a_0,s_1)}\left[r_1+\mathbb{E}_{a_1\sim\pi(a_1|s_1)}\left[\mathbb{E}_{s_2\sim p(s_2|s_1,a_1)}\left[\mathbb{E}_{r_2\sim p(r_2|s_1,a_1,s_2)}\left[\underbrace{\gamma r_2+\gamma^2}_{\text{moved the }\gamma\text{ inside}}V(s_2)\right]\right]\right]\right]\right]\right] \\ &= \mathbb{E}_{a_0\sim\pi(a_0|s_0)}\left[\mathbb{E}_{s_1\sim p(s_1|s_0,a_0)}\left[\mathbb{E}_{r_1\sim p(r_1|s_0,a_0,s_1)}\left[\mathbb{E}_{a_1\sim\pi(a_1|s_1)}\left[\mathbb{E}_{s_2\sim p(s_2|s_1,a_1)}\left[\mathbb{E}_{r_2\sim p(r_2|s_1,a_1,s_2)}\left[\underbrace{r_1}_{\text{moved }r_1\text{ inside}}+\gamma r_2+\gamma^2 V(s_2)\right]\right]\right]\right]\right]\right] \\ &= \mathbb{E}_{a_0\sim\pi(a_0|s_0)}\left[\mathbb{E}_{s_1\sim p(s_1|s_0,a_0)}\left[\mathbb{E}_{r_1\sim p(r_1|s_0,a_0,s_1)}\left[\mathbb{E}_{a_1\sim\pi(a_1|s_1)}\left[\mathbb{E}_{s_2\sim p(s_2|s_1,a_1)}\left[\mathbb{E}_{r_2\sim p(r_2|s_1,a_1,s_2)}\left[r_1+\gamma r_2+\gamma^2+\mathbb{E}[...]\right]\right]\right]\right]\right]\right] \end{aligned}$$

And now you can appreciate why G_0 is usually just denoted as $G_0 = \mathbb{E}\left[\sum_{i=0}\gamma^i r_i\right]$

I was asked to upload a picture of the whiteboard, though I don't think it can be that interesting... anyway, for completeness, this was the whiteboard: